

Sabah Ummie

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Norfolk, VA - 23505, US

RESEARCH INTEREST

AI/ML in Healthcare, Agentic AI, Explainable/Responsible AI, AI Interpretability, AI/ML Analysis of Gene/Disease Data, AI in Drug Discovery.

EDUCATION

- **University of Rhode Island** Jan 2021 - Dec 2022
MS in Electrical and Computer Engineering
◦ GPA: 3.82/4.00
Rhode Island, US
- **North South University** Sep 2007 - Aug 2012
BSc. in Computer Science and Engineering
Dhaka, Bangladesh

RESEARCH EXPERIENCE

- **Agentic LLM Benchmarking for Small Molecule Drug Discovery [Ongoing]**
Researcher at Elite Research Lab
 - Analyze the limitations of traditional LLM evaluation frameworks in interpreting underlying chemical rules in small-molecule drug discovery benchmark.
 - Investigate multi-turn agent evaluation for tool selection accuracy, ReAct reasoning, and context retention over iterative interactions over single-turn LLM benchmarks
 - Analyze multi-agent workflow architectures focusing on task extraction and tool orchestration.
 - Analyze evaluation criteria for the workflow for integrating chemical APIs like Boltz2 and ADMET-AI for autonomous drug discovery.
- **Develop Federated Model Predicting Postpartum Depression Risk [Ongoing]**
Researcher at Elite Research Lab
 - Engineer a decentralized machine learning system utilizing the Flower framework and XGBoost to evaluate Postpartum Depression prediction.
 - Compute the optimal global feature selection, to maximize computational efficiency and achieve a peak predictive accuracy of 77.5%.
 - Implement a collaborative FedXgbBagging strategy to improving generalization and the AUC to 79%, from the 72%.
 - Analyze global model evaluation metrics to assess generalization capabilities, demonstrating a highly stable convergence trajectory with logarithmic loss dropping to roughly 0.39 and achieving a maximum unseen test accuracy of 82.5%.
 - Develop an evaluation pipeline to assess the model's classification that achieved an 88.6% overall AUC, 87.5% precision, and an 82.4% F1 Score.
- **RISE ABOVE: Web-based education portal for People With Epilepsy.**
Graduate Research Assistant at University of Rhode Island
 - Developed RISE ABOVE, a Python Flask application to deliver patient-centered, stress-management modules for reducing internalized stigma in people with epilepsy.
 - Implemented human-subject research protocols for the IRB review process, successfully securing regulatory approval.
 - Analyzed focus group usability survey feedback.

PUBLICATIONS & CONFERENCE

C=CONFERENCE, J=JOURNAL, P=IN PREPARATION, S=IN SUBMISSION, T=THESIS

- [C.1] Md Shahadat Jaman, Md. Maniruzzaman, Amzad Sadik Abid, Kazi Nazmul Hasan, Sabah Ummie, Faisal Ahmed. (2025). **Biologically Informed Feature Selection in Alzheimer's Disease**. In *International Conference on Decision Aid Sciences and Applications*, 2025., pp. 2058-2062. doi: 10.1109/DASA68193.2025.11498968.
- [J.1] Allyson S Goldstein, Elaine T Kiriakopoulos, Geoffrey Tremont, Kunal Mankodiya, Sabah Ummie, Seth Margolis. (2022). **A-156 Development of an Online Intervention to Reduce Internalized Stigma in Epilepsy..** Archives of Clinical Neuropsychology, 2022. <https://doi.org/10.1093/arclin/acac060.156>
- [T.1] Sabah Ummie (2022). **RISE ABOVE: A Patient-Centered Wed-Based Intervention For Reducing Internalized Stigma In People With Epilepsy..** *DigitalCommons@URI*, 2022.

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, SQL
- **CS Fundamentals:** Object Oriented Programming (OOP), Data Structures and Algorithms, Java Collection, Java 8 (Stream API), multithreading, and object-oriented designs.
- **Frontend & Backend Technologies:** HTML, CSS, Javascript, Flask, Spring Boot, REST API, Microservice Architecture, MVC
- **Tools and Servers:** GIT, Github, Maven, Asana, Apache Tomcat
- **OS Platforms:** Windows, Linux
- **Data analysis:** Pandas, Matplotlib, FLower (flwr) federated learning library

AWARDS & GRANTS

- **Blue Skies Innovations to Improve QOL** 2022
Epilepsy Foundation of New England
 - **Amount:** \$75,000
 - **Project:** RISE ABOVE: Web-Based Education Portal for People with Epilepsy
 - **Description:** Secured funding to develop and evaluate a patient-centered web-based intervention to reduce internalized stigma and improving social well-being for individuals with epilepsy.

TEACHING & MENTORSHIP

- **Teaching Assistant** Spring 2022
Department of ECBE (URI)
 - CSC 418: Information & Network Security
 - Assisting the professor prepare course materials and supporting students by helping in their assignments and projects.
 - Discuss answers and explanations of exams and assignments to help students understand them well.
 - Conducting on-to-one sessions to provide students individual help with their assignments and projects. Assisted you in grading assignments, exams and projects.

WORK EXPERIENCE

- **Veracity Software Inc.** Feb 2024 - Jul 2025
Java Software Engineer Iselin, NJ
 - Developed scalable backend services and RESTful APIs using Java 8, Spring Boot, Spring MVC, and microservices.
 - Designed MySQL/PostgreSQL databases with JPA and built automated tests using JUnit and Mockito to ensure high-quality releases.
- **Eitacies Inc.** Feb 2023 - Feb 2024
Java Software Engineer Santa Clara, CA
 - Designed and developed enterprise web applications using Java/J2EE, microservices, RESTful APIs.
 - Managed PostgreSQL databases, schemas, stored procedures, triggers, and SQL queries while applying object-oriented development practices.

REFERENCES

1. **Dr. Kunal Mankodiya**
Associate Professor, Department of Electrical, Computer, & Biomedical Engineering
University of Rhode Island
Email: kunalm@uri.edu
Relationship: MS Thesis Supervisor
2. **Yan (Lindsay) Sun**
Professor, CYPHER Director
University of Rhode Island
Email: yansun@uri.edu
Relationship: Course Instructor